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***B.Tech. Degree IV Semester Special Supplementary Examination
February 2020***

**ME 15-1402 METROLOGY AND INSTRUMENTATION
(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60

**PART A
(Answer *ALL* questions)**

(10 × 2 = 20)

- I. (a) What do you mean by uncertainty in measurement?
- (b) Mention the applications of clearance, interference and transitions fits.
- (c) State why and how wear allowance are considered to gauge design.
- (d) Explain why monochromatic light is used for interferometry work and not the white light.
- (e) Define magnification of a projector.
- (f) Write a short note on spirit level.
- (g) What is static calibration?
- (h) Differentiate between static and dynamic characteristics.
- (i) What is a load cell?
- (j) Explain the fundamental laws of thermocouple.



PART B

(4 × 10 = 40)

- II. (a) Why are standards necessary? Explain "Line standard" and "End standard". (6)
- (b) What is the difference between error and accuracy? (4)
- OR**
- III. (a) Describe the necessity of gauge maker's tolerance in gauge design. (6)
- (b) A 50 mm diameter shaft is made to rotate in the bush. The tolerances for both shaft and bush are 0.050 mm. Determine the dimension of the shaft and bush to give a maximum clearance of 0.075 mm with the hole basis system. (4)
- IV. (a) Draw a neat sketch to illustrate the use of Sine bar for measurement of taper plug gauge and explain it briefly. (5)
- (b) Define the terms "Primary Texture" and "Secondary Texture". It is not possible to produce perfectly smooth surface. Justify the statement. (5)
- OR**
- V. (a) Describe and discuss the working principle and application of a Toolmaker's Microscope. (5)
- (b) Explain the method of checking the height of a component with the help of Optical flat. (5)

(P.T.O.)

- VI. With the help of an illustrative example, explain the main functional elements of an instrument? (10)

OR

- VII. (a) Explain any two methods for correction of spurious inputs in an instrument. (5)
(b) Describe the working of mercury in glass thermometer as a first-order instrument. (5)

- VIII. (a) With a neat sketch explain Total radiation pyrometer. (5)
(b) Differentiate between un-bonded and bonded strain gauge. (5)

OR

- IX. (a) How Gas chromatography is used to measure air pollution? (5)
(b) Explain any instrument which is used for Acoustic measurement. (5)
